

Risk Management System

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CHIEF GENERAL MANAGER
MUTUAL FUNDS DEPARTMENT

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All Mutual Funds Registered with SEBI
Unit Trust of India
Association of Mutual Funds in India

Dear Sirs,
Risk Management System

As you are aware, SEBI (Mutual Funds) Regulations, 1996 prescribe the duties and obligations of asset management companies (AMCs) and trustees. The code of conduct prescribed under the Regulations requires that mutual funds should render at all times high standards of service and exercise due diligence and ensure proper care in their operations.

In furtherance of the above objectives and to protect the interests of investors, certain systems, procedures and practices must be followed by all the mutual funds.

You may recall that vide our letter dated March 29, 2001, all the mutual funds were advised to inform us on the adequacy of their risk management systems. Subsequently, on our advice, AMFI in association with Pricewaterhouse Coopers as a part of Indo-US Financial Institutions Reforms and Expansion Project, has undertaken a detailed study on risk management practices followed by all the mutual funds. They have made certain recommendations to ensure a minimum standard of due diligence or risk management system for all the mutual funds in various areas of their operations like fund management, operations, customer service, marketing and distribution, disaster recovery and business contingency, etc.

Operating Manual for Risk Management System

AMFI and Pricewaterhouse Coopers have prepared an operating manual for risk management system for the mutual funds. A copy of the operating manual has been sent to you by e-mail and is also available on SEBI website under the Mutual Funds Section.

The risk management practices in various areas of operations of mutual funds are covered in the operating manual under three categories: (i) Existing industry practices (ii) Practices to be followed on mandatory basis, and (iii) Best Practices to be followed by all mutual funds. Details are given below:

(i) Existing Industry Practices:

Under each head of risk area, the manual covers the exemplary practices followed by some / most of mutual funds in India. It may be mentioned that though there are exemplary practices covered in this category, the extent and degree of observance of these practices differ among the mutual funds. Therefore, it is desirable that all mutual funds should develop their systems and follow these practices, if not being done at present.

(ii) Practices to be followed on Mandatory Basis

All mutual funds shall follow the practices which have been indicated as mandatory in the operating manual. These are - (i) risk management function should be assigned to compliance officer or internal risk management committee or to an external agency (ii) disaster recovery and business contingency plans, and (iii) mutual funds should take insurance cover against certain risks.

(iii) Best Practices to be followed by Mutual Funds

These are the practices which must be adopted by the mutual funds as a part of their due diligence exercise after considering the size of their operations.

How to Implement the Risk Management System

The mutual funds are advised to follow the following step-by-step approach to implement the risk management system:

1. Identification of observance of each recommendation

The mutual funds shall identify areas of current adherence as well as non-adherence of various risk management practices under each of the aforesaid three categories. They shall examine the areas where development or improvement of systems is required.

After identifying the same, the mutual funds shall review the progress made on implementation of the systems on a monthly basis and place the progress report in periodical meetings of Boards of AMCs and trustees. The mutual funds shall ensure full compliance of all the risk management practices within a period of six months.

2. Review of Progress of implementation:

Boards of AMCs and trustee companies shall review the progress made by their mutual funds with regard to risk management practices and the same shall be reported to SEBI at the time of sending quarterly compliance test reports and half-yearly trustee reports.

For the first two quarters – December 2002 and March 2003 – all mutual funds shall send a detailed report informing how each recommendation in each of three categories has been implemented and whether Boards of AMCs and trustees are satisfied.

3. Review by Internal Auditors

After full implementation of the risk management system, it shall be made a part of internal audit from April 1, 2003 onwards and the auditors shall check on a constant basis about the adequacy of risk management systems. Their reports shall be placed before the Boards of AMCs and trustees who shall make comments on the adequacy of systems in the quarterly and half-yearly compliance reports filed with SEBI.

These guidelines are being issued in accordance with the provisions of Regulation 77 of the SEBI (Mutual Funds) Regulations, 1996.

Yours faithfully,

P.K. NAGPAL

ANNEXURE

OPERATING MANUAL FOR RISK MANAGEMENT

FOR INDIAN MUTUAL FUNDS

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I. INTRODUCTION

Risk management can be defined as the "overall process of identifying and understanding the full spectrum of an organisation's risk and taking informed actions to help it achieve its strategic objectives, reduce the likelihood of failure and decrease the uncertainty of overall business performance".

This document sets out an enterprise-wide risk management framework for a Mutual Fund in India. For the purpose of this document, the term "Mutual Fund" is used colloquially to refer to the whole group of entities that constitute the mutual fund organisation; i.e. the Asset Management Company (including its Board of Directors and employees) and the Board of Trustees. It is not used to refer to a Mutual Fund as per the definition in the SEBI Regulations.

The framework manual is intended to serve as a model which will help Mutual Funds monitor and mitigate all the risks faced by them, and also use risk management to increase value for investors. The risk management practices described are based on current exemplary practices and international best practice as identified in our "Survey of Risk Management Practices in the Indian Mutual Fund Industry", March 2002. It also takes into account the recommendations in the same survey document, duly modified based on feedback from AMFI members and finally approved by the AMFI Board on 3 July 2002. (See *Appendix A*). Some of the recommendations are to be mandated by SEBI (*Appendix A, Part 1*), others will be issued by AMFI as best practice guidelines (*Appendix A, Part 2*). The mandatory and best practice recommendations have been incorporated in the framework manual.

All measures described in the manual have been categorised as follows:

- to be mandated by SEBI (as per *Appendix A, Part 1*)
- recommended best practice (as per *Appendix A, Part 2*)
- existing industry practice (exemplary practices followed by some/ most Mutual Funds in India as identified in the "Survey of Risk Management Practices in the Indian Mutual Fund Industry, March 2002).

II. RISK MANAGEMENT FRAMEWORK OVERVIEW

The risk management framework described in this document covers all aspects of a Mutual Fund's operations. Risks have been broadly categorised into five areas:

- Fund Management
- Operations
- Customer Service
- Sales and Marketing
- Other Business Risks.

Risk management measures have been described for each of these areas across three dimensions: policies and procedures, systems and organisation. Additionally, measures for specific risks in each area have also been described.

1. Policies and Procedures

Risk management is most effective when it follows a top-down approach. In this approach, the senior management of the Mutual Fund is the main center of power and responsibility. Based on various factors like the risk appetite and business strategy of the organisation, the philosophy regarding risk should be developed. This philosophy should then be transmitted throughout the organisation in the form of concrete and detailed policies, procedures and guidelines. The policy and procedures documents should build a framework for the effective and efficient management of the fund and should include:

- Investment Policy, including Risk Philosophy (*existing industry practice*)
- Operating Procedures (*existing industry practice*)
- Compliance Manual (*existing industry practice*)
- Code of Conduct (*existing industry practice*)
- Disaster Recovery and Business Contingency Plan (*to be mandated by SEBI*)
- Reporting Framework (*existing industry practice*)

2. Systems

The establishment of an enterprise-wise integrated systems architecture will substantially reduce operational risk. The systems of a Mutual Fund should include the following applications:

- Integrated front and back office systems for fund management, dealing, trade confirmation and settlement (*recommended best practice*)
- Fund accounting system for calculation of net asset values (NAVs) (*existing industry practice*)
- Unit-holder administration and servicing systems for customer service (*existing industry practice*)
- Financial accounting and reporting system for the AMC (*existing industry practice*)

Systems should ideally be integrated and developed using open platform architecture. They should facilitate straight-through processing and also be capable of generating the necessary reports to monitor and manage risks. Security features such as access control, firewalls and virus protection measures should also be established.

3. Organisation

The Mutual Fund organisation should be designed taking into account the following risk management principles:

- Segregation of front office and back office in the AMC (*existing industry practice*)
- Independent verification of data input (*existing industry practice*)
- Establishment of Committees for Investment, Valuation and Audit (*existing industry practice*)
- Development of a second line for key positions (*existing industry practice*)
- Establishment of a risk management function (*to be mandated by SEBI*)

The responsibility of understanding the risks run by the Mutual Fund and ensuring that they are appropriately managed ultimately rests with the Board of Trustees. The Board of Trustees must approve all the risk management and should delegate to the AMC the responsibility of the day-to-day execution of these policies.

Risk Management Function (To be Mandated by SEBI)

The Mutual Fund should have an independent risk management function consisting of one or more risk managers. This function will be responsible for identifying, evaluating or measuring all risks inherent in a mutual fund organisation, as well as establishing controls to mitigate such risks. The risks include:

- *Fund Management*: volatility in performance, style drift and portfolio concentration, interest rate movements, liquidity issues, credit risk
- *Operations*: deal errors, settlement problems, NAV and fund pricing errors, inaccurate financial reporting, fraud, failure of mission critical systems and infrastructure, obsolete systems
- *Customer Service*: errors in deal processing, other investor services, fraud
- *Marketing and Distribution*: new product development, selling and distribution
- *Other Business Risks*: critical knowledge loss, skills shortage, non-compliance, third party risks.

The function should be separate from fund management and should report to the Chief Executive Officer of the AMC. The function could be carried out in a number of ways:

- As an additional function of an existing employee of the AMC, e.g. the Compliance Officer or Internal Auditor;
- Through a Risk Management Committee;
- Outsourced to an external agency; or
- As the Trustees of the mutual fund may deem fit.

III. FUND MANAGEMENT

Policies and Procedures		
<i>Existing Industry Practice</i> - The Mutual Fund should have a documented investment policy. This should: - articulate its investment strategy and risk philosophy (i.e. its attitude towards risk, and the amount of risk it is willing to take as part of its investment strategy) - define the objectives of the Mutual Fund schemes, asset allocation targets and model portfolios (if used) - define the investment process - define limits and mechanisms for monitoring limits at various levels : asset class, industry sector, security, counterparty (these should be consistent with SEBI Regulations, where applicable) - define exceptions and their monitoring - include an approved list of brokers - provide guidelines for transactions with associate companies and inter-scheme transfers - define investment norms for debt instruments based on credit ratings.		
Systems		
<i>Recommended Best Practice</i> - The Mutual Fund should ideally implement a front office system which is integrated with the back office system. This will facilitate: - setting up of parameters such as asset allocation limits, counter parties, securities, associate classifications and authority levels - straight-through processing (i.e. the flow of order and trade information from the front office to the back office without any manual intervention) to allow one time capture of investment decisions - system check on preset parameters and reporting of breaches e.g. whether investments have been made in permitted securities or limits on deal size, etc. have been exceeded - automatic time-stamping of deals - maker-checker authorisations - exception reporting - monitoring of outstanding confirmations, settlements and payments - access controls and firewall - decision support capabilities and portfolio modelling (e.g. scenario analysis, what-if analysis) - risk-adjusted performance measurement (e.g., Value at Risk (VaR) analysis) - reporting on target vs. actual portfolio.		
Organisation		
<i>Existing Industry Practice</i> - The Mutual Fund should establish an Investment Committee. This committee will be responsible for : - laying down the Mutual Fund's investment policy and philosophy with regard to different asset classes, sectors, counterparties, etc., as defined in the Investment Policy Manual - reviewing performance and positions with regard to the objectives of the schemes - researching and reviewing counterparties and debt issuers with regard to credit risk. - The front office (fund management) and back office functions must be segregated. <i>Recommended Best Practice</i> - The Mutual Fund should ideally have segregated research, portfolio management and dealing teams in the front office. <i>To be Mandated by SEBI</i> - The risk management function should be responsible for risk measurement, management and monitoring.		
<i>Specific Risk Management Measures for Fund Management</i>		
Risks	Impact	Risk Management Measures
1. Volatility in performance	- Inconsistent or low returns leading to loss of investor confidence. - Erosion of assets under management leading to loss of revenue.	
1. Unexpected change in market conditions		<i>Existing Industry Practice</i> - The Mutual Fund should, at the minimum, adhere to all SEBI regulations relating to investment limits. - The Mutual Fund should carry out daily or weekly performance measurement, comparing Mutual Fund performance to a specified benchmark or peer group. - The Trustees should review the portfolio on a quarterly basis as required by SEBI Regulations. <i>Recommended Best Practice</i>

		• Funds should consider using portfolio management tools for risk measurement, in keeping with international trends. These tools should be used to manage risks more effectively, and should be capable of carrying out the following analytics: • Quantification of exposure using measures such as Value at Risk (VaR), duration, and tracking error • Risk adjusted performance measurement using Sharpe Ratios, Treynor Measures and Sortino Ratios • Risk benchmarking, i.e. the exposure arising between the actual managed portfolio and the benchmark portfolio • Stress testing and back testing of exposure calculations.e.g., Sharpe's Ratio, Treynor Measures, beta, FAMA decomposition, VaR, etc). The mutual funds should consider using equity derivatives for hedging and rebalancing. •
1. Quality of investment research, facilities, people, procedures		<i>Existing Industry Practice</i> • The Mutual Fund should document the rationale for an investment decision as required by the SEBI regulations. • Ideally, the Mutual Fund should have a dedicated research team. • The Mutual Fund should hire qualified and experienced portfolio managers, research analysts and dealers with adequate experience in the industry. They should be provided continuous training to understand new products, skills, markets and sectors. • The fund management and research teams should have access to research from multiple sources: both internal and external.
1. Execution of deals at sub optimal prices		<i>Recommended Best Practice</i> • As the industry matures and the volumes increase, the Mutual Fund should consider having a dedicated dealing function. • The Mutual Fund should ideally establish clear guidelines for best execution. Independent verification procedures for all deals should be established. Rates and prices for verification should be obtained from independent sources.
1. "Back to back" transactions in debt securities of associates or associate companies (as defined in the SEBI (Mutual Funds) Regulations, 1996 2. Internal deals between schemes or portfolios 3. Investments in securities issued by associates; purchases of securities owned by associates; sales of securities to associates. 4. Joint ventures with associates		<i>Existing Industry Practice</i> • All SEBI regulations regarding restrictions on associate transactions and investments as well as requirements for disclosure must be adhered to. • All SEBI regulations regarding the execution of the deal at the market price and the documentation of justification for the inter-scheme deal must be adhered to. • The inter-scheme deal should be independently verified by Compliance.
1. Style drift and portfolio concentration	• Inconsistent / low returns vis-à-vis similar schemes in the market leading to low investor confidence. • Non-compliance with SEBI regulations.	<i>Existing Industry Practice</i> • All SEBI regulations with regard to investment limits must be adhered to at the time of making the investment. • The Investment Committee should review the portfolio on a regular basis to ensure compliance with regulations. • The compliance officer should monitor the portfolio and review all exceptions. • The Investment Committee should review on a daily or weekly basis the actual portfolio vis-à-vis the model portfolio. • The Investment Committee should have a stated strategy and action plan for rebalancing the portfolio.

1. Interest rate movements	Inconsistent / low returns vis-à-vis similar schemes in the market	<i>Existing Industry Practice</i> - The Mutual Fund should have daily profit-and-loss reporting at securities and portfolio levels to the Investment Committee. <i>Recommended Best Practice</i> - The Mutual Fund should consider using interest rate derivatives to manage risk and rebalance portfolios.
1. Liquidity issues	- Delays or inability to meet redemptions leading to non-compliance with SEBI regulations - Low investor confidence leading to erosion of assets under management.	<i>Existing Industry Practice</i> - All SEBI regulations with regard to redemption periods for different scheme types must be adhered to. - The Investment Committee should monitor the portfolio on a daily basis and periodically review it to track illiquid assets and take corrective action. - The Mutual Fund should segregate corporate and retail investors and require the corporate investors to give prior notice of redemptions. - The Registrar & Transfer (R&T) agent should promptly inform the Mutual Fund of any large redemption. - The Mutual Fund should use tools such as stress-testing of redemptions on portfolios and asset-liability matching to assess liquidity risks. - With due allowance for the Mutual Fund's particular characteristics, the Mutual Fund's marketing efforts should strive to broaden its base so as to reduce its vulnerability to redemption surges. <i>Recommended Best Practice</i> - Funds should make suitable in-principle arrangements in advance for borrowing to deal with unexpected redemptions, in order to avoid delays and difficulties in resorting to borrowing when the need arises. The borrowing should not exceed SEBI limits of 20% of net assets under management.
1. Credit risk	- The issuer may default on principal / interest - Defaults may lead to low investor confidence and hence erosion of assets under management.	<i>Existing Industry Practice</i> - The Investment Committee should research and review issuers with regard to credit risk. - The Mutual Fund must adhere to all SEBI restrictions regarding investments in rated and unrated debt securities. - The Investment Committee should monitor the ratings of all debt issuers that the Mutual Fund has invested in.

IV. OPERATIONS RISKS

Policies and Procedures <i>To be mandated by SEBI</i> - The Mutual Fund must buy insurance cover against third party losses arising from errors and omissions. Third party liabilities refer to liabilities arising out of financial loss to investors or any other third party, incurred due to errors and omissions of directors, officers, employees, trustees, R&T agents, etc. The level and type of cover should be determined by the Trustees, subject to a minimum level of Rs 5 crores. However, Mutual Funds with assets of less than Rs.100 Cr may take insurance cover for an amount of less than Rs.5 Cr as determined by their trustees. The premium for this cover may be paid for in accordance with SEBI regulations. - Custodians must also buy separate insurance cover for errors and omissions. <i>Existing Industry Practice</i> - The Mutual Fund should purchase insurance to cover first party losses. First party losses are those which impact the insured and include asset based losses (due to natural or unnatural disasters such as fire, flood, burglary, etc.) as well as financial or data losses. They also include losses due to acts of infidelity by employees of the insured and computer based crimes such as hacking or virus attacks that may impact the data of the Mutual Fund. - Compliance should review all trading activities at frequent intervals. - Operating procedures should lay down reconciliation activities and their frequency. - End-of-day broker confirmations with records of deals - End-of-day reconciliation of positions with custodian data - At least once a week complete reconciliation of fund accounting system records with custodian records - Daily reconciliation between Mutual Fund and others (banks, counterparty, etc). - The Mutual Fund should establish a personal trading policy and a code of conduct for employees.
Systems <i>Recommended Best Practice</i> The Mutual Fund should consider implementing integrated front and back office systems which will facilitate:

- - Straight-through processing to allow one-time capture of trade details
 - Pre-trade compliance checking
 - Automatic time-stamping of deals
 - Maker-checker authorisations
 - Exception reporting
 - Generation of deal confirmations
 - Monitoring of outstanding confirmations, settlements and payments
 - Cash management
 - Access controls and firewalls, virus protection and other security functionality such as locking of trade data
 - Integrated reporting across the Mutual Fund.

Existing Industry Practice

- The Mutual Fund should ensure that the fund accounting systems used (in-house or by the fund accountant to whom this activity has been outsourced) facilitate:
- - Validation of NAV calculations
 - Automated and manual price feeds
 - Identification of missing prices
 - Flagging of price variances beyond pre-established tolerance levels.

Organisation

Existing Industry Practice

- The Mutual Fund should segregate duties to ensure that an independent person or department carries out matching of trade confirmations.
- The Mutual Fund should appoint a Valuation Committee which meets periodically to review valuation policies.

Specific Risk Management Measures for Operations

Risks	Impact	Risk Management Measures
1. Deal errors	• Incorrect execution of deals in terms of price, volume or asset class, potentially leading to failure of settlement, financial loss or non-compliance with regulations.	<i>Recommended Best Practice</i> • The back office system should be integrated with the front office system to facilitate: • straight-through processing • one-time capture of trade details • maker-checker authorisations • in-built checks and validations. <i>Existing Industry Practice</i> • The Mutual Fund should not allow dealing through personal cell phones of dealers. • The Mutual Fund should have recording facilities in the dealing room.
1. Settlement problems	• Failure to provide securities or payment for securities leading to financial loss and/or reputation loss. • Unsettled deals can have an adverse impact on portfolio performance.	<i>Existing Industry Practice</i> • The Mutual Fund and its Investment Committee should maintain and proactively monitor credit information on all counterparties and establish counterparty lists and limits to avoid default on settlements. • The Mutual Fund should receive daily position reports from the custodian. • The Mutual Fund should establish service level agreements with custodians, including clauses that protect the fund and its investors from settlement errors by the custodian. • The Mutual Fund should move towards trading on the Negotiated Dealing System (NDS) for government securities. <i>Recommended Best Practice</i> • The back office system should facilitate daily fund projections to ascertain liquidity and settlement requirements.
1. NAV and fund pricing errors	• Incorrect NAV's lead to units being bought and sold at the wrong price and investors may be disadvantaged. • Misleading performance information to the investors. • Non-compliance with regulations.	<i>Existing Industry Practice</i> • The Mutual Fund must adhere to all SEBI regulations relating to valuation norms and daily or weekly disclosure of NAVs. • The Mutual Fund should appoint a Valuation Committee which meets periodically to review valuation policies. • The Mutual Fund should maintain documentation of all NAV procedures and methodologies and ensure that the documentation identifies all elements critical to NAV calculation. • The Mutual Fund should ensure that the fund accounting systems used (in-house or by the fund

		accountant to whom this activity has been outsourced) facilitate: • validation of NAV calculations • automated and manual price feeds • identification of missing prices • flagging of price variances beyond pre-established tolerance levels. • The Mutual Fund should carry out periodic compliance and audit checks on the NAV calculation methodology to ensure accuracy of calculations and their compliance with the regulatory requirements.
1. Inaccurate financial reporting	• Non-compliance with regulations and loss of investor confidence on account of incorrect projection of financial health.	<i>Existing Industry Practice</i> • All financial reporting should be subject to audits by internal and external auditors as well as the compliance officer, at quarterly intervals. • The Trustees should review all financial reporting to ensure transparency and accuracy. • The Mutual Fund should ensure that adequate disclosure is made with regard to derivative transactions, off-balance sheet items and contingent liabilities.
1. Fraud	• Non-compliance with regulations, financial loss, reputation loss.	<i>Existing Industry Practice</i> • The Mutual Fund should adhere to all SEBI regulations and guidelines with regard to personal trading, insider trading and front- running. • The Mutual Fund should establish a personal trading policy and code of conduct, which should be signed off by all employees at regular intervals. • The Mutual Fund should send continuous reminders about personal trading policies and possible action in case of violation of these policies, to its employees. • The Mutual Fund ideally should consider mandating use of a designated broker or its own dealing room for execution of personal trades of employees. • All customer assets must be maintained with the custodian. • The Mutual Fund should periodically provide training and information to all employees on compliance issues and policies on personal trading and fraud.
1. Failure of mission critical systems and infrastructure	• Failure to complete critical end-of-day operations such as NAV calculation, redemptions and settlement of trades leading to non-compliance and loss of reputation.	<i>To be Mandated by SEBI</i> • The Mutual Fund, and its Registrar and Transfer (R&T) agents and custodians, should have an off-site back up facility and a Business Contingency Plan that is tested and evaluated on a regular basis. The business contingency plan should be comprehensive and should cover information technology, infrastructure and personnel requirements. Such a contingency plan should allow the AMC to perform, at the bare minimum, the critical functions of mutual fund operations on "Day 1". These should include: • Calculation of daily NAVs • Redemption processing • Outstanding trade settlements. <i>Existing Industry Practice</i> • The Mutual Fund should address issues of security and virus protection by using firewalls, access controls and appropriate virus control procedures for its systems and servers. • The Mutual Fund should carry out due diligence on all third-party systems before engaging them and ensure that they have appropriate business contingency plans in place.
1. Obsolete systems	• Operational errors, delay in meeting regulatory requirements, inefficient processing of customer related processes.	<i>Existing Industry Practice</i> • The Mutual Fund should carry out a periodic systems audit to ensure required functionality vis-à-vis products and regulatory requirements. • The Mutual Fund should carry out periodic stress testing of systems to ensure the ability to process large volumes at acceptable speeds.

		The Mutual Fund should implement applications that are developed using open architecture in order to facilitate interfacing and integrating with other applications.
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V. CUSTOMER SERVICE

Policies and Procedures		
<i>Existing Industry Practice</i> - The Mutual Fund should define service levels with regard to investors and incorporate these in the service level agreements with the R&T agent. - The Mutual Fund should establish reconciliation procedures with regard to: - matching of cash receipts to issue of units and cash payments to redemption of units - end-of-day reconciliation to ensure correct allotment, transfer, or redemption of units to investors - end-of-day reconciliation to ensure payment of commissions to all agents - periodic reconciliation between the fund accounting system, R&T system and bank account. - The Mutual Fund should purchase insurance to cover customer litigation. - The service level clauses with the R&T agent should include a liability clause in case of errors and omissions. - The Mutual Fund should have documented policies with regard to complaints handling. - The Mutual Fund should conduct a periodic audit of all investor-related activities, carried out both by the Mutual Fund and the R&T agent, to ensure that all allotments, redemptions, income distributions and commission distributions have been accurate and timely. This audit should also include the dispatch of statements and annual reports. <i>To be Mandated by SEBI</i> - The R&T agent should be required to take separate cover for errors and omissions.		
Systems		
<i>Existing Industry Practice</i> - The R&T system should ideally have the following functionality: - imaging and bar-coding of applications - automatic generation of customer confirmations - maker-checker authorizations - workflow based processing - in-built checks on price and units - flagging units that are pledged - interface capabilities with fund accounting system - automatic calculation of income and redemption amounts - automatic calculation of agent commissions with in-built checks for accuracy - automatic printing of confirmations, statements, income and commission vouchers - exception generation and reporting - auto-update of account information - interface capabilities with sales and marketing database - support of customer and account relationship data model - in-built checks to ensure all statements are printed. - The Mutual Fund should ideally establish electronic interfaces its with banking systems to allow for automatic instructions for payment and reconciliation.		
<i>Specific Risk Management Measures for Customer Service</i>		
Risks	Impact	Risk Management Measures
1. Errors in deal processing	Failure to correctly and timely process customer transactions leading to loss of investor confidence and non-compliance.	<i>Existing Industry Practice</i> - The Mutual Fund should establish procedures for accepting applications and sending out end-of-day confirmations for transactions. Procedures should include scanning of all customer applications. The Mutual Fund should also consider tools such as bar-coding, optical character recognition (OCR), intelligent character recognition (ICR) and the use of pre-filled forms. - The R&T system should facilitate maker-checker authorizations. - The R&T system should facilitate reconciliation of cash and units. - The Mutual Fund should establish controls for alternate channels of distribution such as the telephone and Internet, if used. For transactions carried out over the telephone, call scripts should include confirmations of transaction details. Transactions via the Internet will transfer the responsibility of data entry and its accuracy to the investor.
1. Other investor services	- Failure to meet with regulatory standards leading to non-compliance. - Failure to satisfactorily meet investor complaints and queries, to calculate correct income and distribute it on	<i>Existing Industry Practice</i> All SEBI regulations with regard to investor servicing and complaint resolution, and its reporting, must be adhered to.

	time, and to provide comprehensive and accurate financial information leading to loss of investor confidence. Failure to correctly calculate commissions for agents or to make payments to agents leading to loss of agent confidence in the Mutual Fund.	- The complaint resolution system should facilitate: - tracking of complaint resolution - automatic update of complaints log and forwarding of complaints to the compliance officer. - All complaint resolution processes should be controlled by the compliance officer and the complaint log should be regularly reviewed for assessing the quality and timeliness of resolution. - The Mutual Fund should carry out an audit of all income calculations and commission calculations to ensure accuracy of payments. - The Mutual Fund should adopt the practice of direct debits to customers' and agents' bank accounts.
1. Fraud	Financial loss to the Mutual Fund leading to erosion of per unit NAV.	<i>Existing Industry Practice</i> - All investor units should only be issued in dematerialized form. - The Mutual Fund should conduct periodic audits of customer account set-up and credit checks on customer accounts to prevent the setting-up of fraudulent accounts. - In instances where physical unit certificates are issued, the Mutual Fund should consider the use of bar coding, invisible ink or other tools to prevent the possibility of fraudulent certificates being redeemed. <i>Recommended Best Practice</i> - Cash applications should not be permitted - Redemptions should only be made to a bank account For any change in address request, the R&T agent should confirm the change to both the old and new addresses.

VI. MARKETING AND DISTRIBUTION

Specific Risk Management Measures for Marketing and Distribution

Risks	Impact	Risk Management Measures
1. New product development	Non-compliance with regulations, loss of market reputation and poor customer service leading to loss of investor confidence.	<i>Existing Industry Practice</i> - A new scheme should adhere to all required SEBI regulations which require every new scheme to be approved by the Trustees and the Board of Directors, and the offer document to be reviewed by SEBI. - The Mutual Fund should have a new product process in place. - All new schemes have to obtain clearance of the compliance officer and the regulator before they are launched. - Comprehensive market research should be undertaken by the Mutual Fund before the launching of a new product in order to assess the product's viability in the market. - The launch of a new product should have close involvement of the R&T agent and the IT teams, to enable evaluation of all infrastructure for its capacity to handle unexpectedly large volumes generated by a new scheme. Contingency arrangements should be made to handle overflow volumes.
1. Selling and distribution	Financial loss or delayed returns to investors due to mis-selling, loss of reputation to the Mutual Fund.	<i>Existing Industry Practice</i> - The Mutual Fund should continuously pursue training and certification of all its distributors. It should adopt strict policies and guidelines with regard to selling of its products and enforce these policies by getting the distributor to sign a declaration of adherence to a code of conduct. - The compliance officer should vet all marketing material and brochures given to the distributor. - The Mutual Fund should follow AMFI's Three Part Program for Distributors, comprising Training, Certification and Registration. - The Mutual Fund should use the services of only those distributors who have obtained certification. The Mutual Fund should empanel brokers based on specific criteria, including certification criteria. - The Mutual Fund should use techniques such as "mystery shoppers" and investor surveys to evaluate the adherence of distributors to the code of conduct. It should continuously carry out such checks on its distributors.

VII. OTHER BUSINESS RISKS

Policies and Procedures		
<i>Existing Industry Practice</i> - The Mutual Fund should have documented Human Resources (HR) policies and procedures. The policies and procedures should address issues such as: - attracting and retaining key skilled staff - succession planning - career development plan - training plan to equip new employees with relevant skills and to update skills of existing employees; training could be delivered in-house or through external institutions. - The Mutual Fund should have a well documented and comprehensive compliance manual as required by SEBI regulations, and an operations manual, which are accessible to all employees.		
Organisation		
<i>Existing Industry Practice</i> - The Mutual Fund should have qualified HR and administration staff with a sound knowledge of HR. - The Trustees Board composition should have a majority of independent Trustees as specified by the SEBI regulations. - The compliance officer should have a direct reporting line to the Trustees.		
<i>Specific Risk Management Measures for Other Business Risks</i>		
Risks	Impact	Risk Management Measures
1. Critical knowledge loss	Poor performance of the Mutual Fund in the market vis-à-vis other mutual funds.	<i>Existing Industry Practice</i> - The Mutual Fund should have documented HR policies and procedures addressing issues such as attracting and retaining key skilled staff. Incentives such as stock options, performance bonuses and competitive salaries should be considered. - The Mutual Fund should have well documented policies and procedures. - The HR plan should aim to identify and build a second line for key positions. - The HR plan should also cover holiday planning in case of key employees going on leave.
1. Skills shortage	- Low growth and poor performance vis-à-vis other mutual funds in the market. - Lack of knowledgeable personnel in the organisation leading to a lackluster or negative image of the Mutual Fund in the market.	<i>Existing Industry Practice</i> - The Mutual Fund should have a training plan for employees to update their existing skills and equip them with new skills. The training plan should identify both in-house and external institutions training requirements. - The Mutual Fund should ideally have a dedicated knowledge management function within the organization to disseminate knowledge on new products, markets and developments. This function should maintain links to important information sources relevant for the industry.

1. Non-compliance	• Inability to meet with regulatory requirements leading to a loss of reputation and loss of investor confidence.	<i>Existing Industry Practice</i> • The compliance officer should adhere to all SEBI regulations with regard to the immediate reporting to SEBI of any instance of non-compliance by the Mutual Fund as well as periodic reporting to Trustees. • The Mutual Fund should have a well documented and comprehensive compliance manual, as required by SEBI regulations. This manual should be accessible to all employees. • All new employees should be made to undergo compliance training. • Trustees should adhere to all SEBI regulations regarding their responsibilities. • The Independent directors on the Mutual Fund Board and the Audit Committee should actively review functioning of the Mutual Fund. • The Compliance Department of the Mutual Fund should be staffed with people having legal, regulatory, accounting and other required expertise. • The compliance officer should have a direct reporting line to the Trustees. <i>Recommended Best Practice</i> • Trustees should update themselves with the general and specific roles and responsibilities expected of them, including the proposed Risk Management Framework.
1. Third party risks (R&T agent misuses access to client information, R&T agent gets bought by a customer or goes out of business, custodian goes out of business).	• Customer information gets shared with competitors. • Disruption to operations • Financial loss to investors.	<i>Existing Industry Practice</i> • The Mutual Fund should carry out due diligence on R&T agent and custodians before selection. The service level agreements (SLA) should prohibit the misuse of client information. The Mutual Fund should conduct periodic audits of R&T and custodian activities. • The Mutual Fund should periodically review the arrangement with the R&T agent and the custodian, and also survey other service providers in the market.

APPENDIX A: FINAL RECOMMENDATIONS AS APPROVED BY THE AMFI BOARD ON 3 JULY 2002

PART 1: RECOMMENDATIONS TO BE MANDATED BY SEBI

1. Risk Management Function

We recommend that all funds should have an independent risk management function. This function will be responsible for identifying, evaluating or measuring all risks inherent in a mutual fund organisation, as well as establishing controls to mitigate such risks. The risks include:

- Fund Management: volatility in performance, style drift and portfolio concentration, interest rate movements, liquidity issues, credit risk
- Operations: deal errors, settlement problems, NAV and fund pricing errors, inaccurate financial reporting, fraud, failure of mission critical systems and infrastructure, obsolete systems
- Customer Service: errors in deal processing, other investor services, fraud
- Marketing and Distribution: new product development, selling and distribution
- Other Business Risks: critical knowledge loss, skills shortage, non-compliance, third party risks.

The function should be separate from fund management and should report to the Chief Executive Officer of the AMC. The function could be carried out in a number of ways:

- As an additional function of an existing employee of the AMC, e.g. the Compliance Officer or Internal Auditor;
- Through a Risk Management Committee;
- Outsourced to an external agency; or
- As the Trustees of the mutual fund may deem fit.

A Risk Management Framework manual detailing the policies and procedures, systems, organisation controls and specific risk management measures for the above risks will be prepared by AMFI.

The creation of such a function should be mandated by SEBI, with an implementation time frame of 3 months from the date of such mandate, or 1st January 2003, whichever is later.

1. Disaster Recovery and Business Contingency Plans

All funds, and their Registrar and Transfer (R&T) agents and custodians, should have an off-site back up facility and a Business Contingency Plan that is tested and evaluated on a regular basis. The business contingency plan should be comprehensive and should cover information technology, infrastructure and personnel requirements. Such a contingency plan should allow the AMC to perform, at the bare minimum, the critical functions of mutual fund operations on "Day 1". These should include:

- Calculation of daily NAVs
- Redemption processing
- Outstanding trade settlements.

SEBI should mandate the above with an implementation time frame of 6 months from the date of such mandate.

1. Insurance

We recommend that funds should be required to buy insurance cover against third party losses arising from errors and omissions. The level and type of cover should be determined by the Trustees, subject to a minimum level of Rs 5 crores. However, Mutual Funds with assets of less than Rs.100 crores may take insurance cover for an amount of less than Rs.5 crore as determined by their trustees. The premium for this cover may be paid for in accordance with Chapter VII, Section 52 (4) (b) (x) of the Securities and Exchange Board of India (Mutual Fund) Regulations, 1996. R&T agents and custodians should also be required to take separate cover for errors and omissions.

PART 2: RECOMMENDATIONS TO BE ISSUED AS BEST PRACTICE GUIDELINES BY AMFI

These are recommended best practice measures that should be adopted as the industry matures and the operations of individual players grow in size and complexity.

1. Liquidity Risks

2. Use of Risk Measurement Tools in Portfolio Management

Funds should make suitable in-principle arrangements in advance for borrowing to deal with unexpected redemptions, in order to avoid delays and difficulties in resorting to Mutual Funds should consider using one or more of the following portfolio management tools for risk measurement, in keeping with international trends. These tools should be used to manage risks more effectively, and should be capable of carrying out the following analytics:

- Quantification of exposure using measures such as Value at Risk (VaR), duration, and tracking error
- Risk adjusted performance measurement using Sharpe Ratios, Treynor Measures and Sortino Ratios
- Risk benchmarking, i.e. the exposure arising between the actual managed portfolio and the benchmark portfolio
- Stress testing and back testing of exposure calculations.

1. Front Office and Dealing Systems

Funds should consider implementing integrated front and back office systems which will facilitate straight-through processing, thereby reducing the possibility of input errors at any stage in the investment, dealing and settlement process. More importantly, a front office system with a robust compliance module will facilitate pre-trade compliance checks, thereby reducing the possibility of regulatory or internal limits being breached.

2. Dealing and Best Execution

Currently, most players are too small to warrant a segregation of duties between fund managing and dealing. However, as the industry matures and volumes increase, this will be an area that should be looked at more closely, with a view to setting clear guidelines for best execution.

3. Money Laundering

In the absence of any money laundering regulation in India, funds should, at a minimum, adopt the following measures:

- Cash applications should not be permitted
- Redemptions should only be made to a bank account
- For any change in address request, the R&T agent should confirm the change to both the old and new addresses.